

Every failure mode measured on the VCS has a documented neural-network analogue

The map reads as a necessary-condition screen: on the VCS each failure is proven against the exact §3.2 intervention oracle (full observability, no learning); on neural networks the same failure is a documented analogue, not proven here. A method that fails the screen should not be trusted for

Failure mode stronger claim with no other evidence measured on the VCS	VCS measured evidence (necessary, not sufficient) proven in this benchmark	Neural-network analogue documented, not proven here	Status here → there
1 Gradient vanishes on discrete / index outputs	0.000 position-regime F single record Vanilla saliency & IG faithfulness = 0.000 on the position regime; the whole gradient/correlational bucket = 0.068 (n=9). Provably 0 at a strobe-timed argmax readout (§3.2 oracle).	Shattered / saturated gradients; non-differentiable argmax & index readouts (analogue refs G1, G2).	proven here ↓ analogue there
2 Saliency is model-invariant (sanity-check fails)	0.000 sanity-check margin single record Program-randomization test: saliency cosine = 1.000 to the real map while 95% of RAM changed -> sanity check FAILS. Constant Jacobian on the content path, no ReLU.	Saliency unchanged under model- & label-randomization; behaves like an edge detector (analogue ref S1).	proven here ↓ analogue there
3 Correlation ≠ causation (Granger & tuning trap)	0.136 Granger F 95% CI [0.02, 0.29] Granger-edge faithfulness = 0.136 (false edges vs the true data-flow); game-variable tuning = 0.116 (spurious tuning) -- both vs the exact §3.2 oracle.	Spurious Granger edges; 'present ≠ used' tuning & attention-as-explanation (analogue refs C1, C2).	proven here ↓ analogue there
4 Single-unit ablation has an interaction blind-spot	0.414 connectome F 95% CI [0.08, 0.78] Single-unit lesion importance is faithful (0.987) yet MISSES the wiring: the connectome graph is recovered at only 0.414; SAE-feature ablations move y in 0 of the audited cells.	Single-unit ablation misleading; computation distributed across units (analogue refs D1, D2).	proven here ↓ analogue there
5 SAE / probe: present-not-used + polysemanticity	0.162 selectivity F 95% CI [0.08, 0.24] SAEs match a known variable (100% of candidates in 4/6 games, min 83%) yet score 0.195 on causal use; probes decode at 0.894 acc but 25 cells are decodable-not-causal -> selectivity F = 0.162.	Control-task selectivity gap; superposition / polysemantic neurons (analogue refs P1, P2).	proven here ↓ analogue there
6 Baseline-sensitivity of Integrated / Expected Gradients	0.034 EG F (all-regime) 95% CI [0.00, 0.10] IG magnitude is baseline-sensitive (corr_sensitivity up to 0.999 across games); a baseline equal to the content byte kills IG entirely; Expected Gradients lands at 0.034.	Attribution depends on the (arbitrary) baseline choice; no canonical reference input (analogue refs B1, B2).	proven here ↓ analogue there

Headline (position regime): causal/intervention 0.412 vs gradient/correlational 0.068 (95% CI 0.390 / 0.070) → a 0.344 faithfulness gap; activation patching = 1.000 (oracle ceiling) where vanilla saliency = 0.000 (naive-gradient floor).

■ VCS evidence: proven in this benchmark (vs the exact §3.2 intervention oracle)
■ NN analogue: documented in the literature, not proven by this benchmark

Analogue references (documented in the literature, not measured here):
G1 Balduzzi et al. 2017; G2 Sundararajan et al. 2017 (IG). S1 Adebayo et al. 2018. C1 Anand et al. 2019; C2 Jain & Wallace 2019.
D1 Morcos et al. 2018; D2 Leavitt & Morcos 2020. P1 Hewitt & Liang 2019; P2 Elhage et al. 2022. B1 Kindermans et al. 2019; B2 Sturmfels et al. 2020.

All VCS numbers read from committed records (leaderboard.json, leaderboard_ci.csv, faithful_demo.json, guided_backprop / ig_baseline_sweep / sae / linear_probing core summaries); 95% CIs are bootstrap-over-games (P2-R-U)